

Rezolvare — Model de antrenament TS2

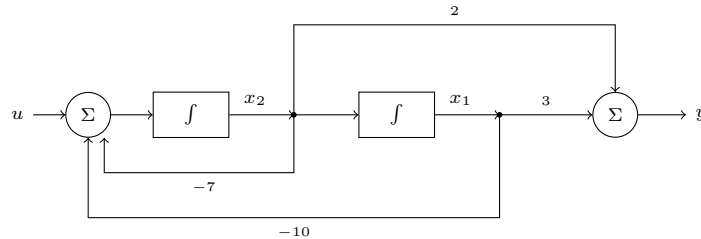
$$1. G_0(s) = \frac{2s + 3}{s^2 + 7s + 10} = \frac{2s + 3}{(s + 2)(s + 5)}$$

a) Model la stare

$$a_1 = 7, a_0 = 10, b_1 = 2, b_0 = 3$$

$$A = \begin{pmatrix} 0 & 1 \\ -10 & -7 \end{pmatrix}, B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, C = (3 \quad 2), D = 0$$

b) Schema bloc



c) Răspuns la treaptă unitară

$$Y(s) = \frac{G_0(s)}{s} = \frac{2s + 3}{s(s + 2)(s + 5)} = \frac{A}{s} + \frac{B}{s + 2} + \frac{C}{s + 5}$$

$$A = \frac{3}{2 \cdot 5} = \frac{3}{10} \quad B = \frac{-4 + 3}{(-2)(3)} = \frac{1}{6} \quad C = \frac{-10 + 3}{(-5)(-3)} = -\frac{7}{15}$$

$$y(t) = \frac{3}{10} + \frac{1}{6}e^{-2t} - \frac{7}{15}e^{-5t}$$

d) Controlabilitate

$$AB = \begin{pmatrix} 1 \\ -7 \end{pmatrix}, \quad P = (B \ AB) = \begin{pmatrix} 0 & 1 \\ 1 & -7 \end{pmatrix}, \quad \det P = -1 \neq 0$$

Controlabil.

$$2. G(s) = 8 \frac{0.25s + 1}{s}$$

a) Bode

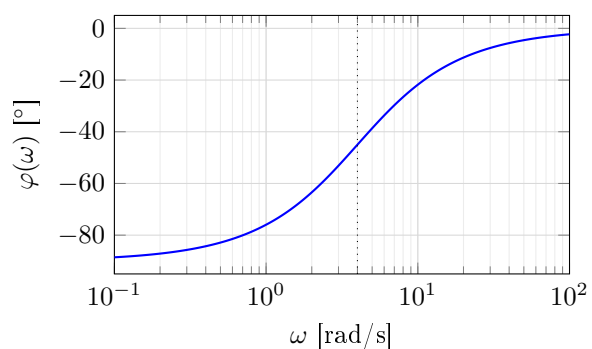
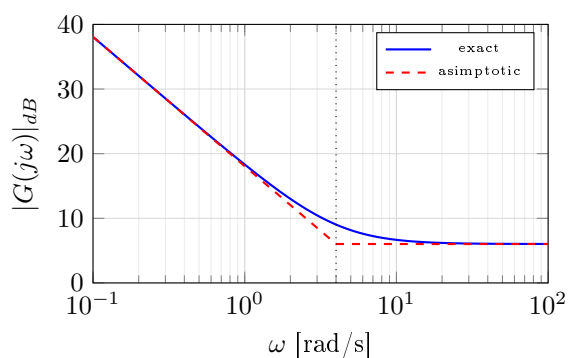
$$k = 8, T = 0.25 \Rightarrow \omega_f = \frac{1}{T} = 4, \quad k_{dB} = 20 \lg 8 = 18.06 \text{ dB}$$

$$|G(j\omega)| = \frac{8\sqrt{1 + (0.25\omega)^2}}{\omega} \quad |G(j\omega)|_{dB} = 18.06 + 20 \lg \sqrt{1 + (0.25\omega)^2} - 20 \lg \omega$$

$$\varphi(\omega) = \arctan(0.25\omega) - 90^\circ$$

Asimptotic: pantă -20 dB/dec pt $\omega < 4$ (trece prin $(\omega=1, 18.06 \text{ dB})$), devine 0 dB/dec pt $\omega > 4$. Valoare asimptotică la $\omega_f=4$: $18.06 - 20 \lg 4 = 6.02 \text{ dB}$ (exact: 9.03 dB , corecție $+3 \text{ dB}$).

$$\varphi(0) = -90^\circ, \quad \varphi(4) = \arctan(1) - 90^\circ = -45^\circ, \quad \varphi(\infty) = 0^\circ$$



b) Erori staționare

$$K_p = \lim_{s \rightarrow 0} G(s) = \infty, \quad K_v = \lim_{s \rightarrow 0} sG(s) = 8, \quad K_a = \lim_{s \rightarrow 0} s^2 G(s) = 0$$

$$e_{ss}(\text{treaptă}) = \frac{1}{1 + K_p} = 0 \quad e_{ss}(\text{rampă}) = \frac{1}{K_v} = \frac{1}{8} = 0.125 \quad e_{ss}(\text{parabolă}) = \frac{1}{K_a} = \infty$$

$$3. \quad G(s) = \frac{k}{s(s+4)}, \quad H(s) = \frac{1}{s+2}$$

$$L(j\omega) = \frac{k}{-6\omega^2 + j\omega(8 - \omega^2)}$$

a) Curba polară & domeniu de stabilitate

$$\text{Im}\{L(j\omega)\} = 0 \Rightarrow \omega(8 - \omega^2) = 0 \Rightarrow \omega_\pi = 2\sqrt{2} \approx 2.828$$

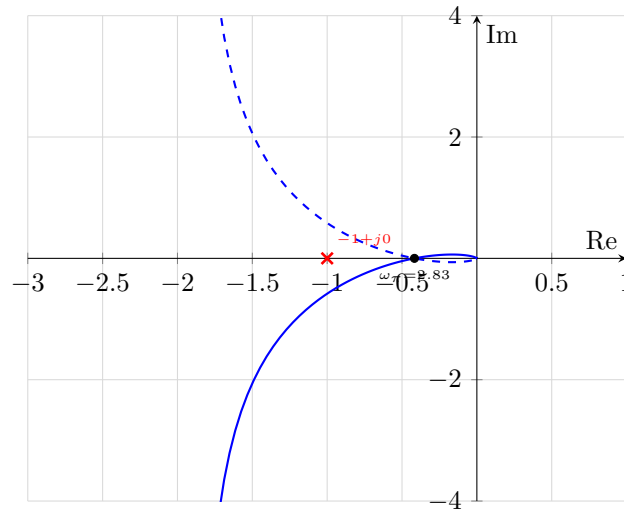
$$\text{Re}\{L(j\omega_\pi)\} = \frac{k}{-6 \cdot 8} = -\frac{k}{48}$$

$$-\frac{k}{48} > -1 \Rightarrow k < 48 \quad \Rightarrow \quad 0 < k < 48$$

$$\text{Re}(\omega) = \frac{-6k\omega^2}{36\omega^4 + \omega^2(8 - \omega^2)^2} \quad \text{Im}(\omega) = \frac{-k\omega(8 - \omega^2)}{36\omega^4 + \omega^2(8 - \omega^2)^2}$$

Puncte de control (ex. $k = 20$):

$\omega \rightarrow 0^+$:	$\text{Re} \rightarrow 0,$	$\text{Im} \rightarrow -\infty \ (\varphi \rightarrow -90^\circ)$
$\omega = 1$:	$\text{Re} = -1.412,$	$\text{Im} = -1.647$
$\omega = \omega_\pi = 2.828$:	$\text{Re} = -0.417,$	$\text{Im} = 0$
$\omega = 5$:	$\text{Re} = -0.101,$	$\text{Im} = 0.057$
$\omega \rightarrow \infty$:	$\text{Re} \rightarrow 0,$	$\text{Im} \rightarrow 0^+ \ (\varphi \rightarrow -270^\circ)$



b) Marginea de fază (ex. $k = 20$)

$$|L(j\omega_t)| = 1 \Rightarrow 36\omega_t^4 + \omega_t^2(8 - \omega_t^2)^2 = k^2 = 400$$

$$x = \omega_t^2: \quad x^3 + 20x^2 + 64x - 400 = 0 \Rightarrow x \approx 3.005 \Rightarrow \omega_t \approx 1.733$$

$$\varphi(\omega_t) = -90^\circ - \arctan \frac{\omega_t}{4} - \arctan \frac{\omega_t}{2} = -90^\circ - 23.42^\circ - 40.94^\circ = -154.36^\circ$$

$$\gamma = 180^\circ + \varphi(\omega_t) = 25.64^\circ$$

$$4. \quad G(s) = \frac{K(s+4)}{s(s+3)(s^2+2s+5)}$$

Poli: 0, -3, $-1 \pm 2j$ ($n = 4$). Zero: -4 ($m = 1$).

Asimptote

$$n - m = 3, \quad \theta_k = \frac{180^\circ(2k+1)}{3} = 60^\circ, 180^\circ, 300^\circ$$

$$\sigma_a = \frac{(0 - 3 - 1 - 1) - (-4)}{3} = \frac{-1}{3} \approx -0.333$$

Unghi de plecare din $-1 + 2j$

Vectori de la ceilalți poli: din $0 \rightarrow (-1+2j)$: 116.57° ; din $-3 \rightarrow (-1+2j)$: 45° ; din $-1-2j \rightarrow (-1+2j)$: 90° (sumă 251.57°). Vector de la zero: din $-4 \rightarrow (-1+2j)$: 33.69° .

$$\varphi_p = 180^\circ - 251.57^\circ + 33.69^\circ = -37.88^\circ \quad (\text{din } -1-2j : +37.88^\circ, \text{ prin simetrie})$$

Puncte de rupere

$$K(s) = -\frac{s(s+3)(s^2+2s+5)}{s+4} = -\frac{s^4+5s^3+11s^2+15s}{s+4}$$

$$\frac{dK}{ds} = -\frac{3s^4+26s^3+71s^2+88s+60}{(s+4)^2} = 0 \Rightarrow 3s^4+26s^3+71s^2+88s+60=0$$

$$s \approx -2.32 \text{ (rupere, pe segmentul } (-3, 0)) \quad \text{și} \quad s \approx -4.85 \text{ (intrare, pe segmentul } (-\infty, -4))$$

Segmente reale ale locului (regulă impar): $(-3, 0)$ și $(-\infty, -4)$.

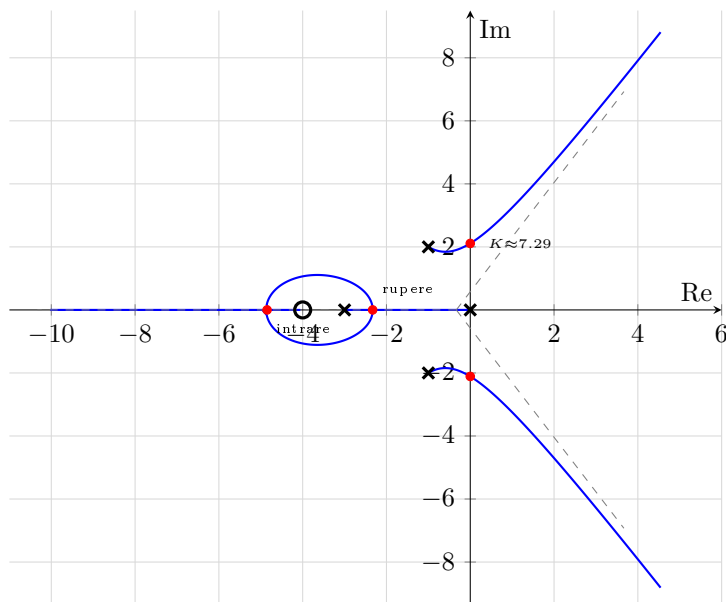
Intersecția cu axa imaginară

Ecuatie caracteristică: $s^4 + 5s^3 + 11s^2 + (15 + K)s + 4K = 0$, cu $s = j\omega$:

$$\text{Re} : \omega^4 - 11\omega^2 + 4K = 0 \quad \text{Im} : \omega(15 + K - 5\omega^2) = 0 \Rightarrow \omega^2 = \frac{15 + K}{5}$$

$$K^2 + 75K - 600 = 0 \Rightarrow K = \frac{-75 + \sqrt{8025}}{2} \approx 7.29$$

$$K \approx 7.29 \quad \omega^2 = \frac{15 + 7.29}{5} \approx 4.458 \Rightarrow \omega \approx \pm 2.11$$



$$5. \quad G(s) = \frac{16}{s^2 + 8s + 16}, \quad u(t) = 3 \sin 3t$$

$$G(j3) = \frac{16}{(16-9) + 8 \cdot 3j} = \frac{16}{7 + 24j}, \quad |7 + 24j| = \sqrt{49 + 576} = 25$$

$$|G(j3)| = \frac{16}{25} = 0.64 \quad \varphi = -\arctan \frac{24}{7} = -73.74^\circ$$

$$y(t) = 3 \cdot 0.64 \sin(3t - 73.74^\circ) = 1.92 \sin(3t - 73.74^\circ)$$

